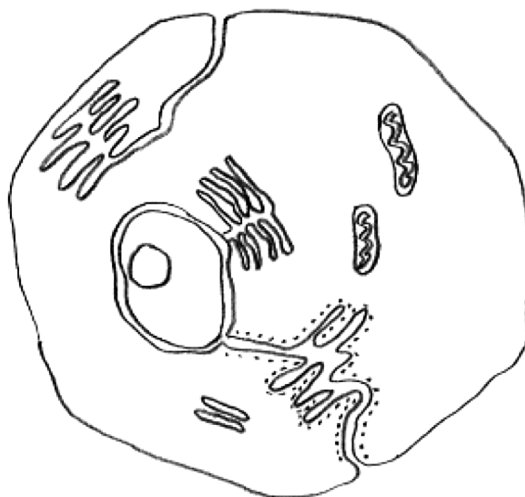


- | | | | | |
|---------------------|------------------------------|----------------------|------------------|-----------------|
| Nucleus | Cell Membrane | Cell Wall | Nucleolus | Ribosome |
| Mitochondria | Endoplasmic Reticulum | Golgi Complex | | Vesicles |
| Centrioles | | | | |



- • •

- G - T - T - A - C - A - G - T

3. (a) Which process of cell division is responsible for the **growth and repair of cells**?
- (b) Which process is involved in producing the **reproductive cells** of an animal?
- (c) What is the meaning of **haploid** and **diploid number**?
4. An organism has 4 chromosomes in its normal body cells. Draw a series of simple diagrams showing the stages involved in **mitosis**. **Name each stage** and state **what is happening** in each.

5. Meiosis occurs in reproductive organs in **two stages**. Draw a series of diagrams to show the stages involved. **Name each stage** and state **what is happening** in each.

6. (a) What is the difference between the male and female sex chromosome pair?
- (b) Who determines the sex of a child - the mother or father? Explain.
7. (a) Show, by Punnett square, why half of any particular population would be expected to be female.
- (b) A family has had five girls born into it, and the mother is pregnant with the sixth child. What is the chance that the child will be a girl? Explain why.
8. Briefly describe the significance of the work of Gregor Mendel to our understanding of genetics.

9. Define the following terms.

allele

pure-bred

hybrid

homozygous

heterozygous

genotype

phenotype

sex-linked

dominant

recessive

co-dominant

incomplete dominance

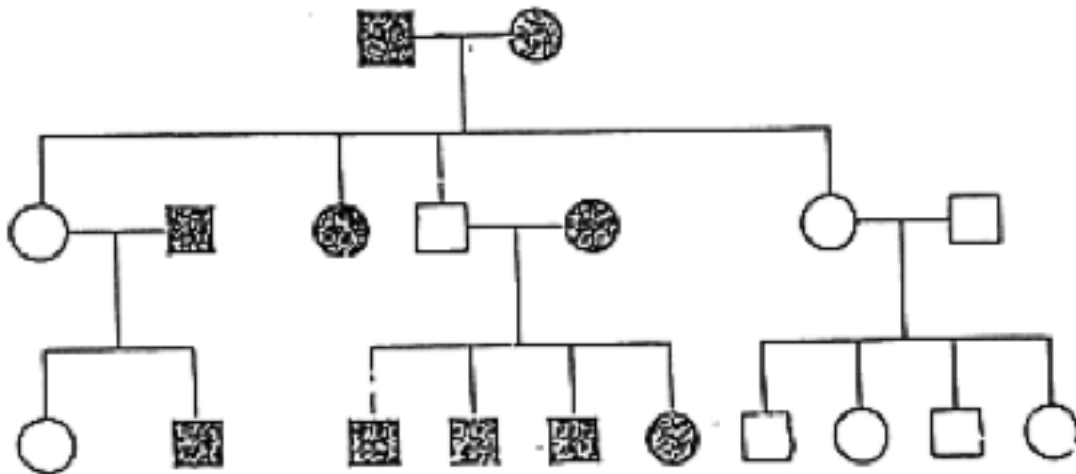
haploid

diploid.

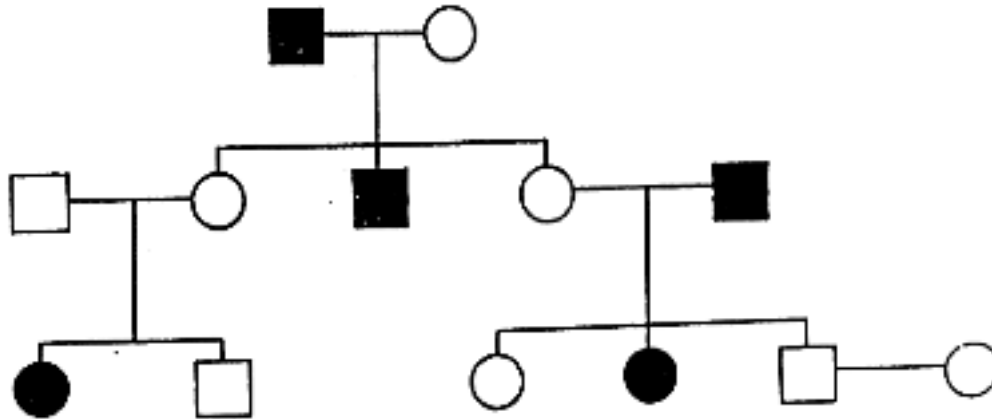
10. Give **one example** of how ***environment*** can affect how a characteristic shows in an individual.

11. In humans, the ability to taste a chemical phenylthiocarbamide (PTU) is due to the presence of the **dominant gene *T***.
- (a) Determine the genotype and phenotypes of the offspring produced by the marriage of a **homozygous non-taster male** and a **heterozygous taster female**.

- (b) Study the pedigree chart below showing the occurrence of tasters in a family. A shaded square or circle represents a taster. Determine the genotypes of every individual.



12. The pedigree below shows the existence of straight hair in a family.



- (a) Is straight hair a dominant or recessive trait? Explain your answer.
- (b) Determine the genotype of all individuals.
13. (a) Explain the difference between **co-dominance** and **incomplete dominance**. You may need to give an example of each to help your explanation.
- (b) In shorthorn cattle, **red coat colour co-dominates with white coat colour**, producing a roan colour in a heterozygous animal. Determine the F_1 and F_2 generation (first and second generation) offspring for a **red bull** mated with a **white cow**. Set out your working.

- (c) In snapdragon flowers, **red colour is incompletely dominant over white**. Hybrid plants have pink flowers. Determine the genotypes and phenotypes for the offspring produced from a **red flowered** plant with a **pink one**.

14. In guinea pigs, **rough coat dominates smooth coat**. Determine the genotypes and phenotypes for the mating of **two rough-coated hybrid** animals.

15. In humans, suppose that the gene that controls right-handedness is dominant over left-handedness. A heterozygous right-handed male and a left-handed female have four children, three of whom are left-handed.
- (a) If R represents the gene for right-handed, use a Punnett square to determine the expected ratio of children that should be right-handed.
- (b) Suggest a reason for difference between the result in part (a) and the actual proportion of children in this family who are right-handed.